Serval Ece Cerdojan
22402843
Section-1

Q1:

0	22	22
1	1	1
2	13	13
3	44	24
4	24	44
5	33	Empty
6	14	Empty
7	29	29
8	Empty	14
9	Empty	33
10	Empty	Empty

Linear Probing

Quadratic Probing

CamScanner ile tarandı

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Section-1

Q2:

a) Initial size is 5

Inserting 7, 12, 3, 18, 25

$N = 5$, $\lambda = 5/5 = 1.0$ No rehash since $N \leq 5$ for all of these keys

Inserting 14

$N = 6$, $\lambda = 6/5 = 1.2 > 1.0$ Needs rehashing

The next prime after 2×5 is 11

Inserting 7, 12, 3, 18, 25, 14 into the new table with size 11

7 mod 11 = 7
12 → 1
3 → 3
18 → 7
25 → 3
14 → 3

Inserting 9, 21, 30, 4, 16 size ≤ 11 so no rehash

Inserting 28 $N = 12 > 11$ so needs rehashing newsize = 23

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
–	–	25	3	4	28	–	7→30	–	9	–	–	12	–	14	–

16	17	18	19	20	21	22
16	–	18	–	–	21	–

– → indicates empty

b) For x to end up at index 4, its hash should be less or equal to 4.

If x is 1 since 1, 2, 3 are full it goes to the 4th index

So minimum x is 1.

CamScanner ile tarandı